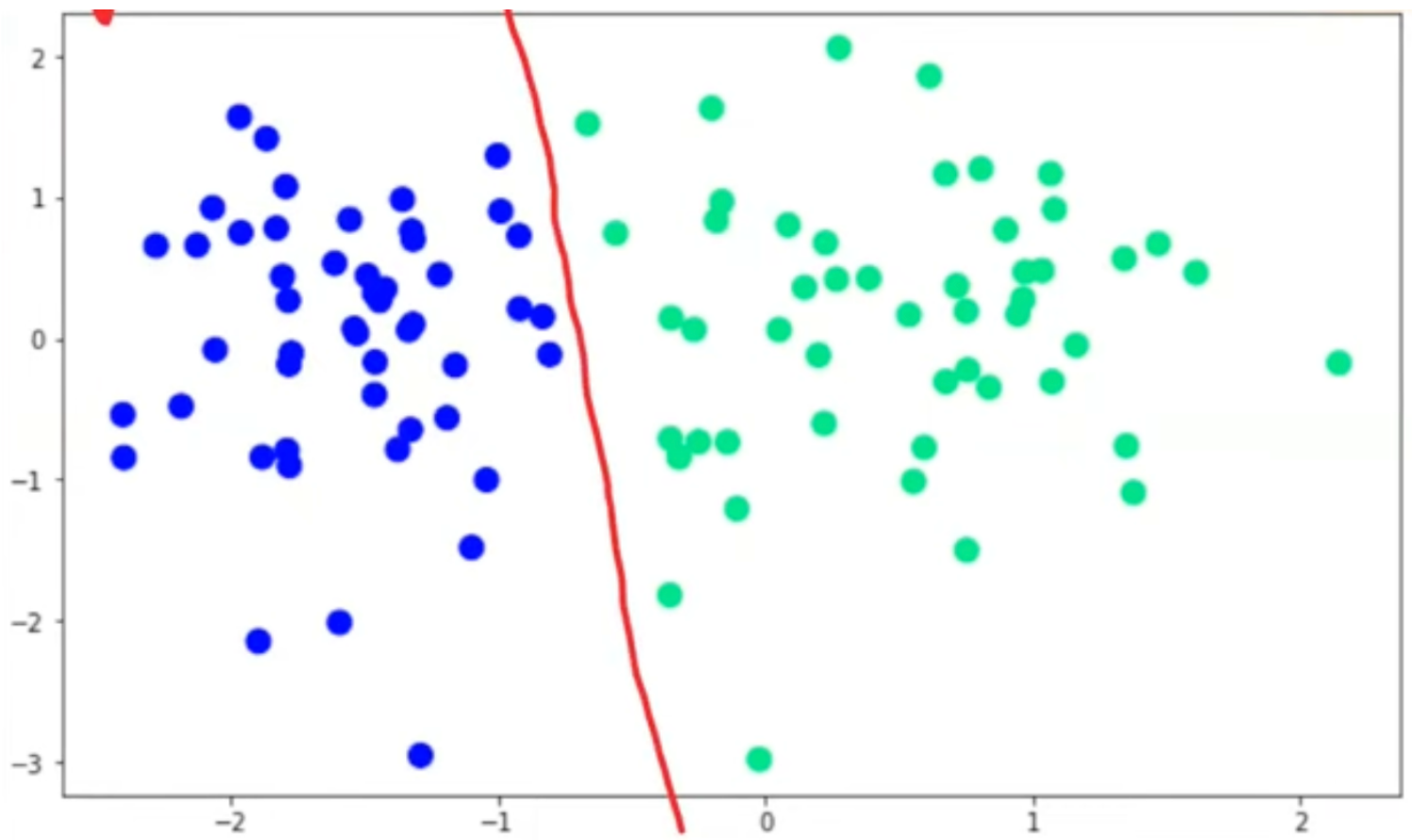


Logistic Regression

Requirements to Use Logistic Regression

1) Data should be almost linearly separable.



Logistic Regression eqⁿ = $AX + BY + c = 0$

* Perceptron Trick

- 1) Start with a random line
- 2) Loop through data points in each epoch
- 3) Done till convergence.

* How label regions

$$Ax + by + c = 0$$

IF $c \uparrow \Rightarrow$ Line goes down

$c \downarrow \Rightarrow$ Line goes up

IF $x \uparrow \Rightarrow$ Line rotates upward

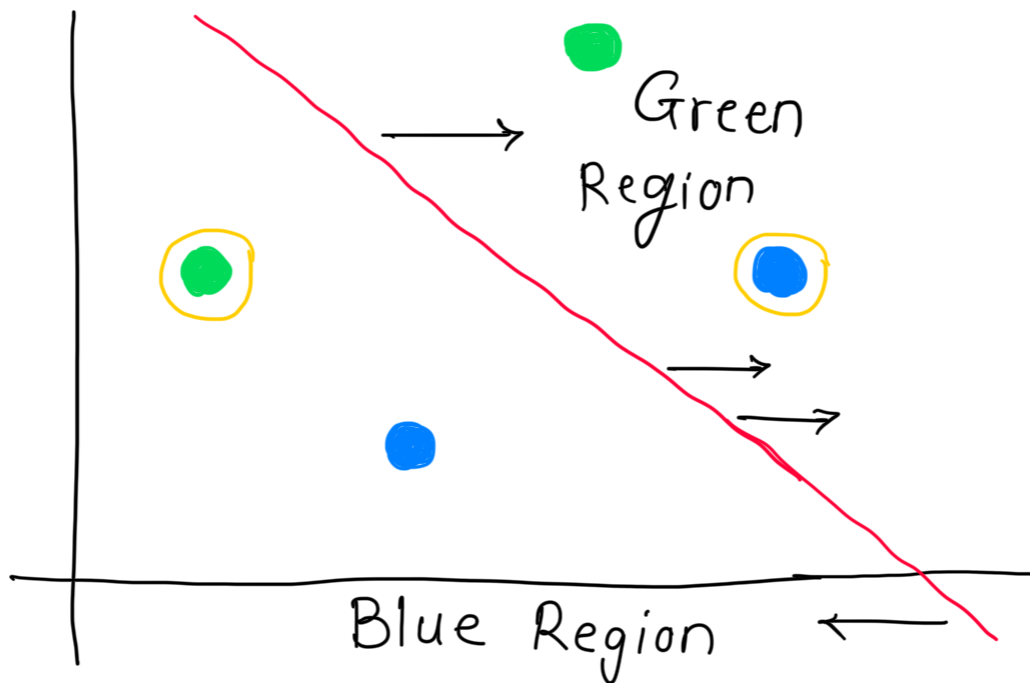
$x \downarrow \Rightarrow$ Line rotates downward

(Anticlockwise)

IF $\gamma \uparrow \Rightarrow$ Line rotates downward

$\gamma \downarrow \Rightarrow$ Line rotates upward
(clockwise)

* Transformation



Highlighted points are misclassified
so we need to move the line in the
direction of blue misclassified pt.

Suppose

$$\text{line eq}^n = 2x + 3y + 5 = 0$$

Point's co-ordinates = (4, 5)

So to move line upward

Perceptron Trick suggest to add
1 as a co-ordinate & then subtract
From lines co-ordinates

i.e.

$$\begin{array}{r} 2 \quad 3 \quad 5 \\ - \quad 4 \quad 5 \quad 1 \\ \hline \end{array}$$

$$-2x - 2y + 4 = 0 \Rightarrow \text{New line}$$

For Green point we have to move
line downward & to do so we
have to add the point co-ordinates
to the lines co-ordinate

$$\begin{array}{r} 2 \quad 3 \quad 5 \\ + \quad 1 \quad 3 \quad 1 \end{array}$$

$$\underline{3x + 6y + 6 = 0 \Rightarrow \text{New line.}}$$

* If +ve point is in -ve region then we add co-ordinates to the co-ordinates of line and vice-versa.

In case of ML, we don't change the direction of line this sudden, instead we use learning rate.

$$\therefore \boxed{\text{coeff} = \text{coeff} - \eta \text{co-ordinate}}$$

* Algorithm

E is read as belongs to

$$x_i \in N \text{ and } \sum w_i x_i \geq 0$$

$$w_n = w_0 - \eta x_i$$

$$x_i \in N \text{ and } \sum w_i x_i < 0$$

$$W_n = W_0 + \eta x_i$$

Simplified

$$W_n = W_0 + \eta (y_i - \hat{y}_i) x_i$$

y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
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1	1	0	$W_n = W_0$
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0	0	0	$W_n = W_0$
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1	0	1	$\Rightarrow x_i \in P$ but $\sum W_i x_i < 0$ $\therefore W_n = W_0 + \eta x_i$
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0	1	-1	$\Rightarrow x_i \in N$ but $\sum W_i x_i > 0$ $\therefore W_n = W_0 - \eta x_i$
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for i in range (epochs = 1000):

select a random student C_i

$$W_n = W_0 + \eta (y_i - \hat{y}_i) x_i$$

IMP

	x_1	x_2	y
X_0	CGPA	IQ	Placed
1	7.5	81	1
1	8.9	109	1
1	7.0	81	0

$Ax + By + c = 0$ can be written

as $w_0 + w_1x_1 + w_2x_2 = 0$

$w_0 = c, w_1 = A, w_2 = B$

$w_0x_0 + w_1x_1 + w_2x_2 = 0$

$\Rightarrow \sum_{i=0}^n w_i x_i = 0 \Rightarrow$ Used in above
eqn